Software as a Medical Device (SaMD) at the FDA: Regulatory Science and Review

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***Abstract* — Development and deployment of software as a medical device (SaMD) have increased in modern times. At the U.S. Food and Drug Administration, we ensure that such medical devices are safe and effective within their intended use cases. This paper reviews the FDA approval process for medical devices and describes several regulatory science efforts within the Division of Imaging, Diagnostics, and Software Reliability to understand and facilitate the regulatory review of computational medical devices.**

Keywords— FDA, Regulation, OSEL, DIDSR, artificial intelligence, deep learning, machine learning, software as a medical device

I. Introduction

As computational medical devices have grown in use in the modern world, so has the need to develop rigorous evaluation methodologies for their uses. Software as a medical device (SaMD) has applications across a wide product area including machine-learning-based medical image analysis and atrial fibrillation detectors. The U.S. Food and Drug Administration (FDA)'s Center for Devices and Radiological Health (CDRH), ensures that marketed medical devices are safe and effective for their intended use.

CDRH protects and promotes the health of the public by ensuring the safety and effectiveness of medical devices and the safety of radiation-emitting electronic products. We provide consumers, patient, their caregivers, and providers with understandable and accessible science-based information about the products we oversee. We facilitate medical device innovation by advancing regulatory science, providing industry with predictable, consistent, transparent, and efficient regulatory pathways, to stimulate consumer confidence in devices marketed in the U.S[1].

The Office of Science and Engineering Laboratories (OSEL) within CDRH conducts laboratory-based regulatory research to facilitate development and innovation of safe and effective medical devices and radiation emitting products. We provide scientific and engineering expertise, data, and analyses to support the regulatory processes, including pre-market review, post-market investigations, and standards development. We also collaborate with academia, industry, government, and standards development organizations to develop, translate, and disseminate science and engineering-based information regarding regulated products. A goal for OSEL is to release to the public regulatory science tools (RST) to aid companies in assessing their medical device safety and effectiveness for pre-market approval[2].

The Division of Imaging, Diagnostics, and Software Reliability (DIDSR) conducts research across a wide variety of imaging and diagnostic devices to develop methods for evaluations including methodologies for characterizing the image quality of emerging imaging systems, characterizing new medial image display devices, dose reduction protentional of new image reconstruction methods, and performance assessment of artificial intelligence and machine learning algorithms. DIDSR also develops state-of-the-art methods for the design of clinical trials involving imaging devices, and evaluation of the resulting trial data to enable more efficient and effective utilization of imaging data and more powerful clinical studies[3]. This paper provides a high-level overview of the medical device premarket review process and a description of the research performed by DIDSR on SaMDs in medical imaging devices.

II. FDA Premarket Process for Medical Devices

Medical devices are quite diverse in their nature and applications and are classified and regulated according to their safety risk to the public[4]. Class I devices contain low-risk devices, such as band-aids, tongue depressors, and medical examination gloves. Class II devices contain moderate-risk devices which could cause patient harm, such as medical imaging devices. Class III devices are high-risk devices where failure may be fatal to the patient, such as cardiac stents.

Table 1 shows a high-level overview of the different device classes and the premarket requirements for such devices. For most low-risk devices classified as Class I, they are exempt from the premarket review processes, but are subject to General Controls. General controls are baseline requirements for the Food, Drug, and Cosmetic Act that apply to all medical devices, Class I, II, and III.

Most Class II devices go through the premarket notification pathway, commonly known as the 510(k). In a 510(k) submission, the entity submitting the medical device for FDA

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approval need to demonstrate substantial equivalence to a different Class II predicate device and meet the Special Controls for such devices[5]. Special Controls are requirements for devices which general controls alone are not sufficient to provide reasonable assurance of safety and effectiveness of the device. Generally, Special Controls are device specific, and examples include premarket data requirements, performance standards and guidelines, special labeling requirement, and post-market surveillance.

Table 1: Device class and premarket requirements

Device Class	**Controls**	**Premarket Review Process**
Class I (lowest risk)	General Controls	Most are exempt
Class II	General Controls Special Controls	Premarket Notification [510(k)]
		De Novo
Class III (highest risk)	General Controls Premarket Approval	Premarket Approval [PMA]

For devices that do not have a legally marketed predicate device, the De Novo Classification Request can be submitted[6]. The De Novo pathway is a request to classify novel medical devices as Class I or Class II using a risk-based assessment, and if the request is granted, the device may be marketed and used as predicates for future premarket notifications (510(k)) submissions, when applicable.

Class III devices are approved under the Premarket Approval (PMA) process[7]. The PMA is the most stringent type of device marketing application required by the FDA and requires a standalone demonstrate of the safety and effectiveness of the device. PMA is used when general and special controls alone are insufficient to assure the safety and effectiveness of the device.

III. Regualtory Science Research in FDA/CDRH/OSEL/DIDSR

Within FDA/CDRH/OSEL, there are multiple regulatory science research that is associated with SaMDs. This section will briefly describe several projects in DIDSR.

A. In Silico Imaging Trial for Breast Imaging

Expensive and lengthy clinical trials can delay regulatory evaluation of innovative technologies, which affects patient access to high-quality medical products. Simulation is increasingly being used in product development, but rarely in regulatory applications. A study by Badano et. al[8], attempted to demonstrate the utility of in silico data for regulatory approval by conducting a computer-simulated imaging trial evaluating digital breast tomosynthesis (DBT) as a replacement for digital mammography (DM) and to compare the results with a comparative (physical) clinical trial. The simulated Virtual Imaging Clinical Trial for Regulatory Evaluation (VICTRE) project obtained images with in-silico anthropomorphic breast phantoms and lesions and Monte Carlo x-ray transport coding to simulate a version of DM and DBT systems. Using a simulated observer for the lesion detection task, the results of the simulated VICTRE trial were consistent with comparative real trial for detection of breast lesions, which shows that in silico, imaging trial could be a source for regulatory evidence for imaging devices.

B. Synthetic Images for Algorithm Training

In general, as the number of training cases for deep learning algorithms increases, overfitting tends to decrease, and performance on the target population tends to improve, with a diminishing improvement after a sufficiently large training data set. Unlike other computer vision tasks where labelled data sets are readily available, well-annotated medical imaging data sets are more limited.

In a study by Cha et. al[9], they evaluated whether using computationally generated synthetic mammograms could be used for training data augmentation for a deep learning breast mass detection algorithm. Using the tools developed for the VICTRE trial, they generated synthetic mammograms that mimic the training data set of a real data set of mammograms, the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM)[10]. On a testing subset of the CBIS-DDSM data set, they observed that in general, the detection performance increased when training with synthetic images compared to training without them, but the amount of improvement depended on the number of real and synthetic images used in training, with the performance increasing yielding a diminishing performance increase as the number of synthetic cases increase. The results of this study showed that computationally generated data sets can be a useful tool for data augmentations in deep-learning breast mass detection algorithms.

C. Assessment of Computer-aided Traiage Devices

A computer-aided triage and notification (CADt) device typically use deep-learning algorithms to prioritize specific radiological medical image reads if such image is suspicious of time-sensitive conditions such as stroke, intercranial hemorrhage, and pneumothorax[11]. The performance of these devices is currently evaluated based on how accurate the algorithm performs in identifying patients with a given time-sensitive condition. However, quantitative assessment of clinical effectiveness of CADt for prioritizing the review of patients with these conditions is unclear.

Thompson et. al[12, 13] present a mathematical framework based on queueing theory to computationally simulate the average wait time per patient for patients with time-sensitive condition with and without the usage of a CADt device. Their model predicts the effectiveness of the wait-time-saving as a function of several parameters, including disease prevalence, patient arrival rate, radiologist reading rate, number of

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radiologist on-site, algorithm performance, and the presence of patients who may override the CADt prioritization. Using a defined performance metric to measure the device's time-saving effectiveness, they show quantitively that a CADt device is more effective in situations simulating busy, short-staffed setting.

D. Deep Learning in Digital Pathology

Digital pathology whole-slide images (WSI) are large, gigapixel images. Computational image analysis of such images using deep-learning algorithms could help pathologists diagnose patients. However, multiple challenges of using digital pathology images need to be addressed, including processing such large images, the difficulty in obtaining image-based annotation, and the need to normalize the image characteristics across multiple vendors of the scanner hardware.

To address the issue of processing the large images in a reasonable time, Li et. al[14] performed parallelization of their deep learning algorithm using the CDRH's high-performance computing (HPC) facilities. However, they found that traditional parallelization techniques and file formats can have significant scaling problems with HPC clusters when dealing with such large data sets. They devised a computational strategy and adopt a file format well-suited for parallel data read and writes to facilitate deployment of a trained deep-learning algorithm on the CAMELYON[15] data set. The developed computational framework allowed for significant reduction of computational time for the deployment of the deep learning algorithm.

Kahaki et. al[16] developed a deep-learning algorithm for predicting treatment response for endometrial cancer. Accurate prediction of response to hormonal treatment would allow for personalized treatment. However, using fully annotated WSI is not feasible due to the large size, and only a portion of the pathology samples representing areas of cancer, and asking pathologists to manually annotate all the samples are prohibitively burdensome. To overcome this issue, they devised a weakly supervised deep learning model. Their results demonstrate the potential for using weakly supervised machine learning models on WSI for predicting response to hormonal treatment of endometrial cancer patients.

In digital pathology, previous studies have shown that WSI scanned at different sources can have different color variations. In the study by Arab et. al[17], they participate in the TiGER challenge [18] for automated assessment of tumor-infiltrating lymphocytes in H&E breast cancer slides. They hypothesized that performing color-normalization across the different data sources may improve the generalizability of a deep learning model for segmentation of tumor-stroma segmentation. They show that the performance of the segmentation model substantially increases after color normalization is performed in cross-validation evaluation. This shows that normalizing the characteristics of image inputs to deep-learning algorithms may help their generalizability across the population.

IV. Conclusion

In this paper, we give a high-level overview of FDA activities that relate to medical imaging devices for which SaMDs are becoming common. The medical device regulation at CDRH is briefly explained, and several computational medical imaging research projects in DIDSR were described.

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